



distant planet, carrying and relaying signals from and to the human astronauts back on planet earth.

With this brave new world, just as we have tremendous positives, just as we can enable a lot of people who would otherwise have a more restricted life, just as we can upgrade all human mental capabilities and replace worn-out biological body parts with shiny new mechanical prostheses—we have a potential goliath of a negative hiding around the corner.

The big question is, can we as humans stay in control? Can we remain the dominant life form on earth, or will we hand over power to a machine species, the likes and nature of which we really do not understand?

I believe it's not our physical prowess that has put humans in the powerful, controlling position we have on earth; rather, it's our intelligence. To put it simply, courtesy our intellectual capabilities, and because human brains can mostly out-think and outwit those of other creatures, we have claimed for ourselves the driving seat on earth. But as machine intelligence is rapidly improved, as computer-based systems become more powerful, how long can human domination last?

It is estimated that within a decade, standalone computers will have more processing power than an individual human brain. Not only that, but they will be networked. And what is the intelligence of a machine-brain network?

Even now, a typical computer can perform more calculations in the blink of an eye than a human brain can perform in a week. By giving such

machines the ability to defend themselves under all conditions, we endow them with survival instinct—and this doesn't have to be programmed in.

As of now, it would be quite impossible to actually 'switch off' the Internet. We humans are far too dependent on it. Many lives revolve around the Internet on a day-to-day basis, and already, our present way of life demands that the Internet remain operative. Humans are, in fact, the machines' defenders!

Just as humans do not listen to sheep or dogs—creatures we feel are less intelligent than ourselves, similarly, intelligent machines are very unlikely to listen to humans now. For instance, if a sheep came into your room today and said “Baaah”, you would not say to it, “OK, yes, I will do what you tell me”. After all, the noises the sheep make are trivial in comparison with human language. Likewise, in the future, intelligent machines will find human noises (speech) trivial.

What is to be done about this? Are we all doomed? Clearly, our only hope is to upgrade our own abilities, to create a partnership between human intelligence and machine intelligence. So, far from being something that would be good to do, linking human and machine brains together is quite simply something we have to do.

“If you can't beat them, join them”, appears to be true here. The decision is yours. Will you be happy remaining a human, part of a sub-species in the years ahead? Or wouldn't you rather be upgraded and have extra memory and extra senses? Wouldn't you like to converge with technology? ■

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